NOMENCLATURE

SETS

Ω_L/Ω_N	Set of real branches/nodes of the system
Ω_S/Ω_F	Set of stochastic scenarios/fault events
Ω_B	Set of block times, with $\Omega_B \subset \Omega_s$
Ω_{PV}	Set of candidate nodes for PV units
Ω_{WT}	Set of candidate nodes for WT units
Ω_{ES}	Set of candidate nodes for ES units
Ω_{SE}/Ω_{SF}	Set of real/fictitious substations nodes
Ω_{CL}	Set of critical-load nodes
Ω_{LS}	Set of switchable lines
Ω_{GF}	Set of candidate nodes for grid-forming inverters, $\Omega_{GF} = \Omega_{PV} \cup \Omega_{WT}$
Ω_R	Set of radiality roots, $\Omega_R = \Omega_{SE} \cup \Omega_{SF}$
Ω_H	Set of fictitious branches, $\Omega_H \subset \Omega_{SF} \times \Omega_{GF}$
Ω_T	Set of all branches used in radiality/restoration, $\Omega_T = \Omega_L \cup \Omega_H$.

PARAMETERS

$C^{gf,fix}$	Fixed investment cost for enabling a node with grid-forming capability
$C^{gf,unit}$	Investment cost per converted grid-forming inverter
C^{pv}	Investment cost per PV unit
C^{wt}	Investment cost per WT unit
C^{es}	Investment cost per ES unit
π_s	Probability of stochastic scenario s
π_f	Probability of fault scenario f
σ_n	Load-shedding cost/importance
P_n^d/Q_n^d	Active/reactive power demand at node n
R_{mn}/X_{mn}	Resistance/reactance of branch mn
Z_{mn}	Impedance magnitude of branch mn
FC_s	Demand factor under scenario s
FS_s	PV generation factor under scenario s
FW_s	WT generation factor under scenario s
S^{pv}	Apparent power rating of one PV unit
S^{wt}	Apparent power rating of one WT unit
P^{es}	Power rating of one ES unit
Δt_s	Duration associated with scenario s
η^+/η^-	Storage charging/discharging coefficients
$\overline{V}/\underline{V}$	Maximum/minimum voltage limits
$\bar{I}_{mn}$	Maximum current limit of branch mn
$\overline{N}^{gf}$	Maximum number of grid-forming inverters that can be installed
$\overline{N}_n^{pv}$	Maximum number of PV inverters installable at candidate node n
$\overline{N}_n^{wt}$	Maximum number of WT inverters installable at candidate node n
$\overline{N}_n^{es}$	Maximum number of ES units installable at candidate node n
κ^{pv}	Virtual inertia coefficient of PV-based GFM units
κ^{wt}	Virtual inertia coefficient of WT-based GFM units
κ^{es}	Virtual inertia coefficient of ES-based GFM units
R^{pv}	PV droop coefficient
R^{wt}	WT droop coefficient
R^{es}	ES droop coefficient
f^0	Nominal system frequency
$ROCOF^{max}$	Maximum admissible ROCOF
Δf^{max}	Maximum admissible frequency deviation
M^{isl}	Big-M upper bound for total served load in scenario s, defined as $\sum_n P_n^d \cdot FC_s$
M^{Heq}	Big-M upper bound for equivalent inertia at root g, defined as $\sum_{n=g}\left(\kappa^{pv} \cdot FS_s \cdot \overline{N}_n^{pv} + \kappa^{wt} \cdot FW_s \cdot \overline{N}_n^{wt} + \kappa^{es} \cdot P^{es} \cdot \overline{N}_n^{es}\right)$
M^{Kp}	Big-M upper bound for equivalent droop stiffness at root g, defined as $\sum_{n=g}\left(\frac{S^{pv} \cdot \overline{N}_n^{pv}}{f^0 \cdot R^{pv}} + \frac{S^{wt} \cdot \overline{N}_n^{wt}}{f^0 \cdot R^{wt}} + \frac{P^{es} \cdot \overline{N}_n^{es}}{f^0 \cdot R^{es}}\right)$

CONTINUOUS VARIABLES

$b_{mn,s,f}$	Slack variable used to relax the voltage-drop equation when branch mn is open
$I_{mn,s,f}^{sqr}$	Squared current magnitude on branch mn
$P_{mn,s,f}$	Active power flow on branch mn
$P_{n,s,f}^{se}$	Active power injected by substation at node n
$P_{n,s,f}^{pv}$	Active power injected by PV at node n
$P_{n,s,f}^{wt}$	Active power injected by WT at node n
$P_{n,s,f}^{es,+}$	Active power discharging by ES at node n
$P_{n,s,f}^{es,-}$	Active power charging by ES at node n
$Q_{mn,s,f}$	Reactive power flow on branch mn
$Q_{n,s,f}^{se}$	Reactive power injected by substation at node n
$Q_{n,s,f}^{pv}$	Reactive power injected by PV at node n
$Q_{n,s,f}^{wt}$	Reactive power injected by WT at node n
$V_{n,s,f}^{sqr}$	Squared voltage magnitude at node n
$D_{mn,s,f}^{+}$	Fictitious demand flow in the positive direction
$D_{mn,s,f}^{-}$	Fictitious demand flow in the negative direction
$D_{g,s,f}^{isl}$	Aggregated served load of the island rooted at g
$P_{g,s,f}^{GFM}$	Instantaneous active power available from GFM-enabled units at root g
$\Delta P_{g,s,f}$	Instantaneous active power deficit in the island rooted at g
$H_{g,s,f}^{eq}$	Equivalent inertia of the island rooted at g
$K_{g,s,f}^{p}$	Equivalent droop stiffness of the island rooted at g

BINARY VARIABLES

x_n^{gf}	Binary decision to enable node n for grid-forming operation
$y_{n,s,f}$	Binary energization/restoration of node n
$z_{mn,s,f}$	Binary status of branch mn
$h_{mn,s,f}^{+}$	Binary directional variable for positive directed flow
$h_{mn,s,f}^{-}$	Binary directional variable for negative directed flow

INTEGER VARIABLES

n_n^{gf}	Number of grid-forming inverter units assigned at node n
x_n^{es}	Number of ES units installed at node n
x_n^{pv}	Number of PV units installed at node n
x_n^{wt}	Number of WT units installed at node n
$\tilde{x}_n^{es,gfm}$	Number of ES units at node n operating with GFM capability
$\tilde{x}_n^{pv,gfm}$	Number of PV units at node n operating with GFM capability
$\tilde{x}_n^{wt,gfm}$	Number of WT units at node n operating with GFM capability